

FULL DOC

Server info –

RDP – root@88.198.40.216:53572

SSH – root@88.198.40.216:53572

Password – 8t2kK3wjm6nv3U

You need to run (First Time) –

1. flask run (Keep on Running)
2. python3 mongo_main.py (This will stop)
3. python3 crawler.py (Keep on Running)

Afterwards –

1. flask run (should be Running)
2. python3 crawler.py (should be Running)

All files –

1. app.py - Flask app for open ownership data hosting locally
2. crawler.py
3. data_inp_ownership.py – downloaded json is formatted and inserted into MongoDB
4. data_inp_sanction.py - downloaded json is formatted and inserted into MongoDB
5. downloader.py – script to download in chunks and unzip that json
6. mongo_main.py – define the mongodb collections
7. schema_dict.py – dictionary of all the schema (have some extra document to handle data)

8. ownership_data (folder) – will be generated by crawler.py, will have all the downloading and unzipping of the bulk data.
9. sanction_data (folder) – will be generated by crawler.py will have all the downloading and unzipping of data from opensanction website.
10. ownershipcrawl.log – logging for open ownership
11. sanctioncrawl.log – logging for open sanction

crawler.py – for running the first time it will create 2 folder (ownership_data, sanction_data) and 2 log file (ownershipcrawl.log, sanctioncrawl.log).

Will crawl <https://www.opensanctions.org/datasets/default/> and localhost(the ownership data holding website, hosted locally) every 6 hour and check if the data of update has changed, if changed then it will call the different functions and download, unzip and insert data into the collections accordingly .

For Schema –

1. See the schema_dict.py
2. <https://www.opensanctions.org/reference/>
3. <https://standard.openownership.org/en/0.3.0/schema/schema-browser.html>

Db – sanction
Collections –
Address
Airplane
Asset
Associate
BankAccount
Company
CryptoWallet
Directorship
Employment
entityStatement
Family
Identification
Interest
Interval
LegalEntity
Membership
Occupancy
Organization
Ownership
ownershipOrControlStatement
Passport
Person
personStatement
Position
Representation
Sanction
Security
Thing
UnknownLink
Value
Vehicle
Vessel

Example Query -

```
sanction> db.Person.aggregate([ { $match: {"properties.name": 'Alex Male' } }, { $lookup: { from:
"personStatement", localField: "properties.name", foreignField: "names.fullName", as:
"combined" } } ] );
[
  {
    _id: ObjectId("6526b363d3f992b5c4e44767"),
```

```

id: 'gb-nca-alex-male',
caption: 'Alex Male',
schema: 'Person',
properties: {
  gender: [ 'male' ],
  topics: [ 'crime' ],
  notes: [
    'Male has a number of distinctive tattoos including a skull on his adams apple and a full
sleeve on his left arm with a family tree. 'Love' is tattooed on his left hand along with a swallow
and he has three stars tattooed on his left wrist. He also has a full tribal sleeve tattoo on his right
arm going onto his chest with an angel on his back and 'live' tattooed on his right hand.',
    'Alleged to be a regional distributor of drugs across the south west of England, buying and
selling drugs and managing other distributors who operated under his instruction using the
EncroChat encrypted communications network. He is accused of conspiracy to supply cocaine
and ketamine, money laundering, conspiracy to acquire a firearm.'
  ],
  sourceUrl: [
    'https://www.nationalcrimeagency.gov.uk/most-wanted/alex-male'
  ],
  name: [ 'Alex Male' ],
  ethnicity: [ 'White' ]
},
referents: [],
datasets: [ 'gb_nca_most_wanted' ],
first_seen: '2023-02-13T12:05:50',
last_seen: '2023-10-11T12:02:25',
last_change: '2023-07-03T12:03:00',
target: true,
combined: []
}
]

```

```

sanction> db.Address.aggregate([{$match: {"properties.country": ['ru']}},{ $lookup: { from:
"Identification",localField: "properties.country",foreignField: "properties.country", as:
"combined"}},{ $lookup: { from: "Sanction", localField: "properties.country", foreignField:
"properties.country", as: "combined"} }]);
{
  _id: ObjectId("6526b400d3f992b5c4e6e3e1"),
  id: 'interpol-red-ffdd61cc6a3ee4660d42f0b1d763d3d8410332f3',
  caption: 'Red Notice',
  schema: 'Sanction',
  properties: {
    authority: [ 'INTERPOL' ],
    authorityId: [ '2016/8481' ],

```

```

    sourceUrl: [
      'https://www.interpol.int/How-we-work/Notices/View-Red-Notices'
    ],
    country: [ 'ru' ],
    entity: [ 'interpol-red-2016-8481' ],
    program: [ 'Red Notice' ],
    reason: [ 'participation in illegal armed formation' ]
  },
  referents: [],
  datasets: [ 'interpol_red_notices' ],
  first_seen: '2023-06-06T16:26:42',
  last_seen: '2023-10-11T12:01:24',
  last_change: '2023-06-06T16:26:42',
  target: false
},
{
  _id: ObjectId("6526b400d3f992b5c4e6e3e3"),
  id: 'interpol-red-ffed15885e41a2794ffb9e15dd8f90c808770fd2',
  caption: 'Red Notice',
  schema: 'Sanction',
  properties: {
    program: [ 'Red Notice' ],
    entity: [ 'interpol-red-2016-58117' ],
    sourceUrl: [
      'https://www.interpol.int/How-we-work/Notices/View-Red-Notices'
    ],
    country: [ 'ru' ],
    reason: [ 'Participation in an illegal armed formation' ],
    authorityId: [ '2016/58117' ],
    authority: [ 'INTERPOL' ]
  },
  referents: [],
  datasets: [ 'interpol_red_notices' ],
  first_seen: '2023-06-06T16:26:42',
  last_seen: '2023-10-11T12:01:24',
  last_change: '2023-06-06T16:26:42',
  target: false
}
]
}
]

```

Type "it" for more

please document every Flask route in app.py, why they are here/what they used for
<https://github.com/openownership/bodsdata/tree/main/bodsdataweb>

This will fulfill all the requirements :)

pip install git+<https://github.com/openownership/bodsdata.git@main#egg=bodsdata>

They all are here to replicate the same <https://bods-data.openownership.org/>

document why there's folder named "bootswatch"? No need for that we can remove that. Also you can remove the __init__.py, and the gitignore file

document static and templates folders Static only contains the general bootstrap css. Template contains the all the pages made by open ownership

App.py Why ? app.py replicates the website <https://bods-data.openownership.org/> locally on the server, so that in future if they add 403 forbidden for crawlers we can bypass it. I wanted to do the same for the Open Sanction but they don't have something like this, hope you understand :) Also I was advised to do the same.

Please document process of merging two datasets together These 2 different datasets are not merged as they can't be, they are placed in the same database (MongoDB) to make the query part easy. Also query examples are added into the earlier doc. You can't merge the datasets as they aren't even similar, they are 100% different.

Document making RDF of the data This would require time as I have never made RDF, have heard of it but never tried it , if you want rdf I can provide you ftm's RDF

<https://service.tib.eu/webvowl/#iri=https://alephdata.github.io/followthemoney/ns/ftm.xml>

how to install crawler into cron job

No need to setup the cron job as the script need to run 24x7 and it will automatically crawl every 6 hour.

For setting up the scripts on a new server —

1. Check if mongodb is installed
2. Check the python version
3. Download and unzip the provided script's zip
4. After that cd to that folder
5. Install the python deps by -
`pip install git+https://github.com/openownership/bodsdata.git@main#egg=bodsdata`
6. Then, run `python3 mongo_main.py`
7. Then, run `flask run`
8. Then, run `python3 crawler.py`

If you need you can verify the document count in the mongo shell

If in case the crawler script is found stopped you can simply restart all the scripts —
(also you need to check which scripts has stopped)

1. Then, run `flask run`
2. Then, `rm -r ownerhip_data && rm -r sanction_data`
3. Then, run `python3 crawler.py`